

The values of arcsin a, arccos a, arctan a and arccot a

Undoing a trigonometric function — which needs a restricted domain, and therefore a convention.

LESSON 73–74

2 × 40 MIN

ALGEBRA 10, §4.2

P1 · 5.3

LESSON CLOCK

14'

22'

24'

20'

16'

8'

Why a restriction is needed	14 min
The four ranges	22 min
Standard values	24 min
One value or many?	20 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Explain why the trigonometric functions must be restricted to be invertible.
- State the principal-value ranges of the four inverse functions.
- Evaluate the inverse functions at the standard values.
- Distinguish “the” inverse value from all the solutions of an equation.

TEXTBOOK

National	pp. 313–324
Cambridge	Pure Mathematics 1, pp. 105–110
Workbook	P1 Exercise 5C

01

Explanation

Why a restriction is needed

$\sin x = 0.5$ has infinitely many solutions: $30^\circ, 150^\circ, 390^\circ, \dots$. So the sine is many-to-one, and by the rule of Grade 10 Quarter I it has **no** inverse — until its domain is cut down to a piece on which it is one-to-one.

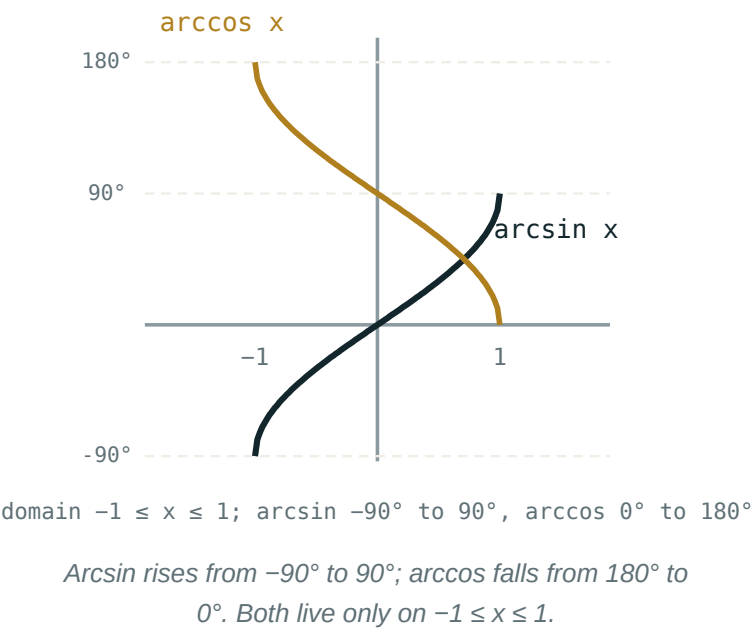
THE CONVENTION

Restrict the sine to $[-90^\circ, 90^\circ]$, where it increases from -1 to 1 exactly once. On that piece it is one-to-one, and its inverse is called **arcsin**.

The choice is a convention, but a universal one: every calculator and every textbook uses it, so an answer outside the range is simply wrong.

The four ranges

FUNCTION	DOMAIN	RANGE (PRINCIPAL VALUES)
$\arcsin a$	$-1 \leq a \leq 1$	$-90^\circ \leq y \leq 90^\circ$
$\arccos a$	$-1 \leq a \leq 1$	$0^\circ \leq y \leq 180^\circ$
$\arctan a$	$\mathbb{R}$	$-90^\circ < y < 90^\circ$
$\operatorname{arccot} a$	$\mathbb{R}$	$0^\circ < y < 180^\circ$



ARCCOS IS NEVER NEGATIVE

$\arccos(-0.5) = 120^\circ$, not -60° . The arcsine's range straddles zero; the arccosine's does not. Confusing the two is the standard error.

Standard values

A	$ARCSIN A$	$ARCCOS A$	$ARCTAN A$
0	0°	90°	0°
$\frac{1}{2}$	30°	60°	—
$\frac{\sqrt{2}}{2}$	45°	45°	—
$\frac{\sqrt{3}}{2}$	60°	30°	—
1	90°	0°	45°
$\sqrt{3}$	—	—	60°
$-\frac{1}{2}$	-30°	120°	—

Two identities are worth knowing, and both follow from the ranges:

$$\arcsin(-a) = -\arcsin a \quad \arccos(-a) = 180^\circ - \arccos a$$

$$\arcsin a + \arccos a = 90^\circ$$

One value or many?

TWO DIFFERENT QUESTIONS

“**Evaluate** $\arcsin 0.5$ ” has one answer: 30° . “**Solve** $\sin x = 0.5$ ” has infinitely many: $30^\circ + 360^\circ k$ and $150^\circ + 360^\circ k$. The first is a function value; the second is an equation.

Lesson 79–81 of Quarter IV is entirely about turning the first into the second. For now the distinction is the whole point: $\arcsin$ gives you **one** angle, chosen by convention, and the other solutions must be produced separately.

02 Worked examples

EXAMPLE 1 Evaluate $\arcsin(-\frac{\sqrt{2}}{2})$, $\arccos(-\frac{1}{2})$ and $\arctan(-1)$.

① $\arcsin \frac{\sqrt{2}}{2} = 45^\circ$, and arcsine is odd.
 -45°

② $\arccos \frac{1}{2} = 60^\circ$; use $180^\circ - 60^\circ$.
 120°

③ $\arctan 1 = 45^\circ$, and arctangent is odd.
 -45°

ANSWER

$-45^\circ, 120^\circ, -45^\circ$

EXAMPLE 2 Evaluate $\sin(\arccos 0.6)$.

① Let $\theta = \arccos 0.6$, so $\cos \theta = 0.6$ and $0 \leq \theta \leq 180^\circ$.

② $\sin^2 \theta = 1 - 0.36 = 0.64$

③ $\sin \theta = 0.8$
 Positive, since θ is in the first two quadrants.

ANSWER

0.8

EXAMPLE 3 Why is $\arccos(-0.7)$ not -45.6° ?

- ① The range of $\arccos$ is $[0^\circ, 180^\circ]$.
- ② A negative value is impossible.
- ③ $\arccos(-0.7) = 180^\circ - \arccos 0.7 \approx 134.4^\circ$

ANSWER

It lies outside the range; the answer is $\approx 134.4^\circ$

03 **Interactive model**

Draw the restricted piece of each curve before evaluating anything.

Principal values*arcsin 1:* 0° 45° 90° 180° *arccos 0:* 0° 90° 180° -90° *arccos(-0.5):* -60° 60° 120° 240°

$\arcsin(-0.5)$:

-30°

30°

210°

330°

The range of $\arctan$ is:

$[0^\circ, 180^\circ]$

$(-90^\circ, 90^\circ)$

$[-90^\circ, 90^\circ]$

$\mathbb{R}$

$\arcsin a + \arccos a$ equals:

0°

90°

180°

it depends

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Inverse trigonometric function	Teskari trigonometrik funksiya	Обратная тригонометрическая функция
Arcsine	Arksinus	Арксинус
Arccosine	Arkkosinus	Арккосинус
Arctangent	Arktangens	Арктангенс
Arccotangent	Arkkotangens	Арккотангенс
Principal value	Bosh qiymat	Главное значение
Restricted domain	Cheklangan soha	Ограниченная область
One-to-one	Biror-birga	Взаимно однозначное
Standard angle	Standart burchak	Табличный угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function needs a restricted domain to be inverted because it must be:

The range of $\arcsin$ is:

$\arccos(-a)$ equals:

“Evaluate $\arcsin 0.5$ ” has:

no answer

one answer

two answers

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\arcsin 0$

ANSWER 0°

2. $\arcsin 1$

ANSWER 90°

3. $\arccos 1$

ANSWER 0°

4. $\arccos 0$

ANSWER 90°

5. $\arctan 0$

ANSWER 0°

6. $\arctan 1$

ANSWER 45°

7. $\arcsin \frac{1}{2}$

ANSWER 30°

Medium · the standard question

7 PROBLEMS

1. $\arccos \frac{1}{2}$

ANSWER 60°

2. $\arcsin(-\frac{\sqrt{2}}{2})$

ANSWER -45°

3. $\arccos(-\frac{1}{2})$

ANSWER 120°

4. $\arctan(-1)$

ANSWER -45°

5. $\arctan \sqrt{3}$

ANSWER 60°

6. $\arcsin 0.6 + \arccos 0.6$

ANSWER 90°

7. $\arccos(-1)$

ANSWER 180°

Hard · stretch and reason

7 PROBLEMS

1. $\sin(\arccos 0.6)$

ANSWER 0.8

2. $\cos(\arcsin \frac{3}{5})$

ANSWER $\frac{4}{5}$

3. $\tan(\arcsin \frac{5}{13})$

ANSWER $\frac{5}{12}$

4. $\arccos(\cos 300^\circ)$

ANSWER 60°

5. $\arcsin(\sin 200^\circ)$

ANSWER -20°

6. Prove $\arcsin a + \arccos a = 90^\circ$

ANSWER Complementary angles in a right triangle

7. Why is $\arcsin(\sin 200^\circ) \neq 200^\circ$?

ANSWER 200° is outside the range of arcsin**07 Homework****Homework — 5 tasks**

Check every answer lies inside the correct range before writing it.

1. Evaluate $\arcsin \frac{\sqrt{3}}{2}$, $\arccos(-\frac{\sqrt{2}}{2})$, $\arctan(-\sqrt{3})$.

2. Evaluate $\sin(\arccos 0.8)$ and $\cos(\arcsin \frac{5}{13})$.

3. Evaluate $\arccos(\cos 250^\circ)$ and $\arcsin(\sin 160^\circ)$.
4. Explain in three sentences why the sine must be restricted before it can be inverted.
5. State the four principal-value ranges from memory and sketch the arcsine and arccosine curves.